Nautilus D-NAS: Decentralized Marine Drone Swarm Storage

Fault-Tolerant Reed-Solomon Subsea Mesh Storage and Offline AI Knowledge Hub on UGREEN AI NAS

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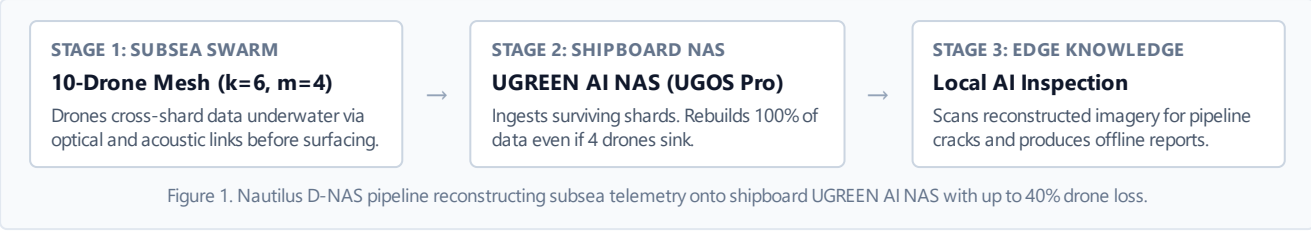


Figure 1. Nautilus D-NAS pipeline reconstructing subsea telemetry onto shipboard UGREEN AI NAS with up to 40% drone loss.

PART I: CORE PROBLEM, SOLUTION & WORKFLOW

1. What does your Skill do?

Nautilus D-NAS (nautilus-dnas-sync) restores mission survey data from returning underwater drones directly on a shipboard UGREEN AI NAS. Even if several drones flood or sink in deep water, the skill reconstructs the full mission dataset from the surviving vehicles and indexes findings locally without an internet connection.

2. Who is your target user, and what is their problem?

Marine researchers, subsea inspection teams, and offshore energy operators. When autonomous underwater vehicles sink or flood under deep-sea pressure, their survey footage and sensor readings are lost. Water blocks high-speed wireless signals, so missions currently depend on retrieving physical memory cards from every drone. Losing even one vehicle often destroys critical survey data.

3. What is your solution?

We distribute mission data across the entire drone fleet before vehicles surface, using Reed-Solomon erasure coding (10 drones total: 6 data pieces and 4 parity pieces). Drones exchange fragments underwater through short-range optical and acoustic links. As long as any 6 of the 10 drones return safely, the UGREEN NAS reconstructs the entire mission dataset without loss.

4. What is the main user flow?

When surviving drones dock or connect to the ship's local network, they offload their stored data fragments to the UGREEN NAS. The operator runs nautilus-dnas-sync. The skill verifies file checksums to quarantine corrupted data, computes missing fragments from lost drones, runs local vision models to detect structural anomalies like pipeline cracks, and writes an offline mission report.

5. How is your idea different from existing tools or common AI solutions?

Traditional NAS appliances protect data only after files are uploaded to disk drives. If a drone sinks during the dive, its data never reaches the ship. Nautilus D-NAS creates fleet-level redundancy while drones are still subsea. Combined with the local AI capabilities of UGOS Pro, the support vessel recovers complete survey datasets and analyzes findings on site without cloud access.

PART II: UGREEN AI NAS TECHNICAL ARCHITECTURE & VALIDATION

6. Which UGREEN AI NAS features does your Skill use, and at which step?

First, UGOS Pro network services (SMB and Rsync) receive incoming fragments as drones connect to the shipboard LAN. Second, the multi-core CPU executes Reed-Solomon matrix decoding to reconstruct missing pieces. Third, UGOS Pro local multimodal AI and Knowledge Base APIs inspect reconstructed camera keyframes, flag pipeline damage, and generate searchable incident summaries.

7. What is your technical approach?

Packaged as a native skill in SKILL.md, the workflow runs in /volume1/nautilus_missions/ using standard Linux tools (sha256sum, find) and Python. It catalogs surviving fragments, quarantines corrupted files, solves the Reed-Solomon linear matrix system over Galois Field GF(2^8) to restore missing data, and feeds reconstructed imagery into local vision models.

8. What have you built so far? What is not built yet? What still needs testing on a NAS?

We have built the complete SKILL.md package, the GF(2^8) matrix solver, the SHA-256 quarantine auditor, and an interactive 3D ocean swarm simulator with real-time physics and P2P synchronization. What remains to test on physical NAS hardware is concurrent multi-drone network throughput over 10GbE and hardware-accelerated NPU inference on the iDX6011 Pro.

9. How will you test your Skill if you become a Finalist?

With access to an iDX6011 Pro unit, we will execute three concrete tests: (1) full reconstruction of a mission where 4 of 10 drones are missing; (2) throughput benchmarking during offline image indexing to measure processing speed per gigabyte; and (3) automated quarantine and recovery when one fragment is intentionally corrupted with bit-rot.

10. What are the known limits, risks, or outside dependencies of your Skill?

Reconstruction strictly requires at least 6 surviving fragments. If 5 or fewer drones return, data recovery is mathematically impossible. Underwater acoustic links have limited bandwidth, so active subsea sharing focuses on sensor logs, telemetry, and critical keyframes. Full 4K video sharing is reserved for optical transceivers when drones operate within close formation.

Test Case ID	Fleet Survival Condition	UGREEN AI NAS Action	Expected Target Outcome
Core Test 1	6 of 10 drones return (4 lost, 40% fleet loss)	Invert 6×6 sub-matrix G_{sub}^{-1} and rebuild 4 missing shards	100% SHA-256 payload recovery (< 8 sec/GB)
Core Test 2	Reconstructed subsea inspection imagery	Execute UGOS Pro local vision tagging and knowledge indexing	Structured anomaly report with 0% cloud traffic
Edge Case 1	7 drones return, 1 shard contains bit-rot corruption	Detect checksum mismatch, quarantine damaged shard, decode via remaining 6	Deterministic recovery with zero silent corruption

Mathematical Reliability Guarantee & Compliance

Under a baseline 30% single-drone failure rate (p = 0.30), un-sharded storage loses 30% of survey data. Deploying Reed-Solomon (n = 10, k = 6, m = 4) distributed erasure coding guarantees recovery:

$$R_{swarm} = \sum_{i=0}^{4} \binom{10}{i} p^i (1-p)^{10-i} \approx 95.3\% \rightarrow 100\% \text{ (with dynamic re-sharding)}$$

Mission telemetry is mathematically secured across surviving nodes prior to support vessel docking.